Page 1: Title and Overview. Section 1: Introduction to Grafting.

Grafting is a horticultural technique whereby tissues of plants are joined so as to continue their growth together.

The upper part of the combined plant is called the scion while the lower part is called the rootstock.

Page 2: Biological Mechanisms. Section 2: Vascular Cambium.

For successful grafting to occur, the vascular cambium tissues of the rootstock and scion must be placed in contact with each o

Both tissues must be kept alive until the graft has taken, usually a period of several weeks.

Page 3: Whip and Tongue Grafting. Section 3: Whip-and-Tongue Grafting.

Method 1: Whip-and-tongue grafting is one of the oldest methods of grafting.

It is commonly used for grafting fruit trees such as apples and pears where scion and rootstock have equal diameter.

Page 4: Cleft Grafting. Section 4: Cleft Grafting.

Method 2: Cleft grafting is used when the rootstock is much larger than the scion.

The stock is split down the center and two scions are inserted into the cleft.

Page 5: Bark Grafting. Section 5: Bark Grafting.

Method 3: Bark grafting, also known as crown grafting, is used for topworking mature trees when the bark is slipping in early spr

Page 6: Bridge Grafting. Section 6: Bridge Grafting.

Method 4: Bridge grafting is utilized to repair damaged trunks where girdling has destroyed the phloem and cambium layer.

Page 7: Side-Veneer Grafting. Section 7: Side-Veneer Grafting.

Method 5: Side-veneer grafting is a widely used technique for propagating conifers and small deciduous nursery stock.

Page 8: Splice Grafting. Section 8: Splice Grafting.

Method 6: Splice grafting is a simple technique where simple diagonal cuts are made on both scion and stock and taped togeth

Page 9: Saddle Grafting. Section 9: Saddle Grafting.

Method 7: Saddle grafting is performed by cutting a V-notch in the rootstock and shaping the scion into a matching wedge or sa

Page 10: Approach Grafting. Section 10: Approach Grafting.

Method 8: Approach grafting involves joining two independent rooted plants together until a union forms, after which the top of t

Page 11: Environmental Conditions and Optimal Temperature. Section 11: Environmental Factors a

The optimal healing temperature for grafting callus formation is 21C to 27C (70F to 80F).

Humidity must be maintained above 85% to prevent desiccation.

Page 12: Tools and Materials. Section 12: Grafting Equipment.

Tools required include sharp grafting knives, budding strips, pruning shears, and grafting wax to seal the wound.

Page 13: Aftercare and Maintenance. Section 13: Post-Graft Care.

Inspect grafts weekly.

Remove suckers emerging from the rootstock below the graft union to direct vigor into the scion.

Page 14: Common Causes of Failure. Section 14: Troubleshooting Graft Failure.

Failures typically result from poor cambial contact, desiccation of tissues, grafting during incorrect physiological seasons, or pat

Page 15: Summary and Glossary. Section 15: Reference Glossary.

Scion: the detached shoot or bud.

Rootstock: the rooted plant part providing root system.

Graft union: the junction where cambium tissues fuse.